

Homework 2

Sonia Kangaju & Abbey Skinner

2026-03-03

Submission + Formatting Requirements

delete this section before turning in

Your submission should include:

- A written report addressing all parts above
- All relevant tables and figures embedded in your report
- Your SAS and/or R code as an appendix
- Selected output (not all raw output—select the relevant portions)

- Write as if preparing a report for a statistical consulting client
- Be specific with your interpretations—relate findings back to your actual variables and research context
- Round test statistics and p-values appropriately (e.g., $F = 4.23$, $p = 0.018$)
- For very small p-values, report as $p < 0.001$

Descriptive Analysis

1. *Descriptive statistics (Abbey)*: Create a table presenting the mean and standard deviation of your continuous outcome at each time point, separately by group.

Systolic Blood Pressure (mmHg) by Period and Sex

Period	Female		Male	
	Mean	SD	Mean	SD
1	130.9	22.1	129.0	16.9
2	136.5	23.2	134.6	19.1
3	140.9	24.2	139.3	21.1

2. *Interaction plot (Sonia)* : Create a plot showing the mean response over time for each group. Based on your plot, comment on whether the profiles appear parallel and whether the groups appear to differ in their overall level

Univariate Repeated Measures ANOVA - Abbey

1. *Sphericity assessment (10 points)*: Assess whether the sphericity assumption is appropriate for your data. Describe your findings and explain how they affect your approach to the within- subject tests.

We used Mauchly's Test of Sphericity which tests if type H covariance structure is valid. After running this test in SAS, we got a p-value of $< .0001$ which tells us that this assumption is not valid for our dataset. To account for this, we will use the adjusted p-values for the within-subject tests.

Sphericity Tests				
Variables	DF	Mauchly's Criterion	Chi-Square	Pr > ChiSq
Transformed Variates	2	0.5907012	1686.2033	<.0001
Orthogonal Components	2	0.9542173	150.10502	<.0001

Figure 1: Mauchly's Test SAS output

2. *Hypothesis tests (20 points)*: Test and interpret the following effects: - Time effect: Is there evidence that the mean response changes over time? - Group effect: Is there evidence that groups differ in their overall mean response? - Group $\times$ Time interaction: Is there evidence that the groups have non-parallel profiles?

For each test, report the relevant test statistic and p-value, and interpret the result in the context of your research question.

For the test of a time effect, we have an F-value of 516.83 and an adjusted p-value of $< .0001$. We have statistically significant evidence that there is relationship between systolic blood pressure over time. This suggests that the lines are not horizontal (have a slope).

For the test of a sex effect, we have an F-value of 7.09 and a p-value of 0.0078. We have statistically significant evidence that there is a relationship between systolic blood pressure and sex. This suggests that males and females have distinct lines (not overlapping).

For the test of an interaction effect, we have an F-value of 0.19 and an adjusted p-value of 0.8179. We do not have evidence that there is an interaction effect between sex and time with relation to systolic blood pressure. This suggests that the lines for sex are parallel.

The GLM Procedure					
Repeated Measures Analysis of Variance					
Tests of Hypotheses for Between Subjects Effects					
Source	DF	Type III SS	Mean Square	F Value	Pr > F
SEX	1	7676.911	7676.911	7.09	0.0078
Error	3204	3470173.534	1083.075		

Figure 2: Repeated Measures ANOVA SAS output, sex effect

MANOVA - Sonia

1. *Test of parallelism (10 points)*: Test whether the group profiles are parallel using MANOVA. State what you found and what the result means for your research question.

2. *Test of group effect (10 points)*: If appropriate based on your parallelism results, test whether the groups have coincident profiles (i.e., same overall level). Explain why testing for coincident profiles is or is not appropriate given your parallelism results.

3. *Test of time effect (10 points)*: Test whether there is an overall time effect using MANOVA. Compare the MANOVA time effect result to the univariate time effect result from Part 2.

The GLM Procedure Repeated Measures Analysis of Variance Univariate Tests of Hypotheses for Within Subject Effects							
Source	DF	Type III SS	Mean Square	F Value	Pr > F	Adj Pr > F	
						G - G	H-F-L
sysbp	2	159794.2686	79897.1343	516.83	<.0001	<.0001	<.0001
sysbp*SEX	2	58.5408	29.2704	0.19	0.8275	0.8179	0.8181
Error(sysbp)	6408	990608.9511	154.5894				

Figure 3: Repeated Measures ANOVA SAS output, time and interaction effects

Comparison and Interpretation

1. *Comparison of approaches (Abbey)*: Compare the results from the univariate and MANOVA approaches. Did they lead to the same conclusions? If not, explain why the results may differ and which approach you consider more appropriate for your data.
2. *Scientific interpretation (Sonia)*: Summarize your findings as you would for a collaborator. What did you learn about how your outcome changes over time and whether groups differ?
3. *Limitations (Both)*: Discuss at least two limitations of these classical approaches for your dataset.

Appendix

```

proc import datafile="/home/u63963979/bios767/data/framingham_wide.csv"
    out=work.framingham
    dbms=csv
    replace;
    getnames=yes; /* Uses first row as variable names */
    guessingrows=max; /* Scans all rows to define column types */
run;

proc glm data = framingham;
    class sex;
    model sysbp1 sysbp2 sysbp3 = sex / nouni;
    repeated sysbp / printe nom;
run;

```

Figure 4: SAS Code for Univariate Repeated Measures ANOVA